MILITARY SPECIFICATION

CORRECTOR, TEMPERATURE: 10511080

1. SCOPE

1.1 This specification covers one Temperature Corrector which is used in sound ranging to convert time intervals obtained at ambient temperature to equivalent time intervals at 10° Centigrade.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10511080	Corrector, Temperature

PACKAGING DATA SHEET

10511080	Corrector, Temperature

FSC 1220

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The temperature corrector shall be manufactured in accordance with Drawing F10511080 and drawings pertaining thereto.

3.2 General Specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The temperature corrector shall show no evidence of physical failure and shall meet all requirements of 3.4.1 at room temperature (plus 60 degrees to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.4 Accuracy.-

3.4.1 Correction readings.- With the movable arm aligned with the time interval listed in Table I, the correction (seconds) indicated for the corresponding temperature shall agree with that listed in Table I.

TABLE I

Time Interval (On Curved Scale)	Temperature (On Movable Arm)	Correction Seconds (Vertical Scale)
.2 sec.	-30°C	.0150 ± .001 sec.
.2 sec.	20°C	.0035 ± .001 sec.
1.14 sec.	0°C	.0203 ± .001 sec.
1.14 sec.	60°C	.0965 ± .001 sec.
2.12 sec.	-35°C	.1758 ± .001 sec.
2.12 sec.	25°C	.0554 ± .001 sec.
2.75 sec.	30°C	.0954 ± .001 sec.
2.75 sec.	55°C	.2104 ± .001 sec.
2.9 sec.	-5°C	.0780 ± .001 sec.
2.9 sec.	40°C	.1496 ± .001 sec.

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3.5 Operability.- Arm shall rotate smoothly over base without binding or excessive looseness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 temperature correctors manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

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4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordnace with MIL-STD-105 and the sampling plans specified in Tables II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II

CLASSIFICATION	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Correction readings	3.4.1	4.6.2
102. Operability	3.5	4.6.3
MINOR: NONE		

NOTE 1: The tests in Table II shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three temperature correctors shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Drawings-Visual
302.	General specifications	3.2	MIL-F-13926 - Visual

NOTE 2: The tests in Table III shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Environmental tests.- Three temperature correctors shall be selected at random from each 50 produced, or from each month's production,

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whichever occurs first. Each sample shall meet the requirement and test in Table IV and shall then meet the requirement in Table II, Characteristic 101.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures	3.3.1	4.6.1

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- When the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The temperature correctors shall be exposed to the temperature conditions specified in 3.3.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The temperature correctors shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note 3) to -80°F and shall remain at this temperature for a minimum of one hour. At the completion of this one hour period, the temperature shall then be gradually raised to +160°F and the temperature correctors shall remain at this

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temperarure for a minimum of one hour. At the completion of this one hour period the temperature of the test chamber shall be gredually reduced to room ambient temperature (+60° to +90°F) at which time the temperature correctors shall be removed from the test chamber. The temperature correctors shall then be subjected to a visual and tactile examination to determine that there is no evidence of physical failure. The temperature correctors shall then be subjected to all the tests specified herein to determine overall compliance with 3.3.1. Caution shall be exercised during this test to avoid subjecting the temperature correctors to thermal shock.

Note 3: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling test of 4.6.1.

4.6.2 Correction readings.- The accuracy of the corrections readings shall be visually verified with the aid of an optical device of at least 5 power. The accuracy of the temperature corrector scale shall be checked against the settings specified in Table I. The readings noted shall be accurate within the tolerances specified in Table I of 3.4.1.

4.6.3 Operability.- The arm shall move smoothly without bind or looseness when normal thumb pressure is applied to move the arm. The arm shall be moved several times back and forth through its mode of travel to determine conformance with the requirements of 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10511080. The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use.- The Corrector, Temperature is a component of Plotting Set, Sound Ranging, M53. The table on the back of the corrector is for use in the calculator method of processing time intervals.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of proper levels of preservation, packaging, and packing.

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

(e) Provisions for First Article testing.

Custodian:
Army - MU

Preparing activity:
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Project No. 1220-A227

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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